



create a skill that uses web search (what ever the harness has access to or duck duck go api as a backup). It is a prospecting agent. You define what you want, it uses the pal protocol to identify and extract intent, then find an create signals, search for them, find companies on LinkedIn, etc, then , find the right contacts at each company via LinkedIn search or other, then, it goes ahead and searches for syntax and if not found it goes and guesses syntax, then does an mx record search for the guesses finally it sends it through a free verification system. to get any lead contact you want from signals to email

Created the draft **Signal-to-Email Prospecting** skill package. It turns a prospecting brief into a bounded workflow: PAL intent extraction → signal research → company discovery → relevant contacts → email-pattern research → labeled guesses → DNS checks → approval-gated free verification.

Classification: reusable research skill with tool adapters—not a persistent autonomous agent. A PAL prospecting agent can invoke it.

Package contents

The attached ZIP includes:

File	Purpose
SKILL.md	Complete procedure, input/output contracts, permissions, failure handling, and quality checks
manifest-entry.yaml	PAL routing and registration metadata
plan.json	Template for intent, ICP, signals, limits, and tool mappings
leads.json	Template for companies, contacts, email candidates, evidence, and verification
validate_output.py	Structural output validator
cases.yaml	Test scenarios covering success, catch-all, quota exhaustion, denied approval, and restricted sources

File	Purpose
README.md	Installation, integration, assumptions, and change log

The lead-template structural check passed. Live search, DNS, verification, and harness integration have not been executed.

Workflow behavior

1. Extract the offer, desired outcome, ICP, geography, roles, exclusions, and buying-signal criteria.
2. Convert those criteria into searchable signal hypotheses—then require evidence before labeling a signal observed.
3. Resolve matching companies and their official domains.
4. Find relevant contacts through permitted search, company websites, announcements, and authorized professional sources.
5. Search for published business emails and supported company naming patterns.
6. Generate a limited number of explicitly labeled guesses when published evidence is unavailable.
7. Check DNS once per domain, then submit eligible addresses to the configured verifier after exact-payload approval.
8. Export local lead records with provenance, verification status, unresolved issues, and source evidence.

The skill follows the existing project’s brief-parsing and reversible-default approach while remaining separate from `pal-website-agent.SKILL.md`, which builds websites and agent runtimes. The complete canonical prospecting PAL protocol was not available in the retrieved excerpt, so the package labels its fallback intent mapping as provisional.^[1]

Important implementation distinctions

Requested capability	Packaged behavior
Harness search with DuckDuckGo backup	Prefer the harness search tool; allow an approved full-search DuckDuckGo adapter. DuckDuckGo’s Instant Answer API is not a full search-results API. ^[2]
LinkedIn company/contact discovery	Support permitted search discovery and authorized access; prohibit LinkedIn scraping, bots, and access-control bypass. Prefer independent company sources for extracted facts. ^[3]
MX checks for guessed emails	Check the domain, not each guessed mailbox. DNS routing does not establish mailbox existence; null MX means the domain accepts no mail, while absent MX can have an A/AAAA fallback. ^[4]
Free email verification	Use Hunter as a configurable default candidate, subject to account access and remaining free quota. Official pages show differing allowances, so the skill does not hardcode a free monthly count. ^[5] ^[6]
“Verified” lead	Keep contact identity, email provenance, and verifier result separate. A guessed address remains inferred even when the verifier reports valid; accept-all remains unresolved. Hunter exposes distinct verification statuses. ^[7]

Installation and defaults

Copy the package into your harness’s configured skills directory, register the manifest, and bind its logical capabilities to the harness’s actual tools and schemas. The package supplies instructions and contracts—not live search, DNS, or verifier adapters.

Reversible defaults:

- 10 companies and up to 2 contacts per company
- 90-day signal freshness window
- Up to 3 email candidates per contact
- 60 search queries and a 30-minute run limit
- \$0 spend and at most 20 verification attempts, constrained by remaining free quota

Third-party verification requires approval of the exact provider payload. Outreach, CRM writes, purchases, and deployment stay outside this skill’s authority.

The remaining configuration decisions are your actual harness, approved PAL protocol, full-search backup, verifier account, suppression source, and retention policy. The draft returns honest partial results when those capabilities are missing rather than claiming it can find or verify every contact.



this is a good fire e one to add: <https://emails-checker.net/bulk-email-checker> and build the complete agent skill using python and pal protocol with automated web search and structured lead scoring and create the pal protocol protecting workflow

Built the updated PAL Prospecting Agent v0.2.0 package with Python orchestration, automated-search hooks, structured lead scoring, the PAL protocol, and a protected prospecting workflow. The attached ZIP includes a runnable offline demo; live research requires binding the adapter to your harness’s model and tools.

Classification: agent wrapper + reusable research skill + Python runtime + tool-adapter contracts.

What’s included

Artifact	Purpose
SKILL.md	Complete prospecting procedure, contracts, permissions, and quality gates
agent.yaml and soul.md	Agent identity, routing, tools, and boundaries

Artifact	Purpose
manifest-entry.yaml	Skill registration
prospect.py	Python search orchestration, extraction calls, email candidates, DNS checks, scoring, and exports
PAL-PROTOCOL.md	Intent extraction and protected execution workflow
SCORING.md	Explicit 100-point review-priority rubric
ADAPTER-CONTRACT.md	Contracts for your harness's search, model, approval, and verification tools
Demo adapter and tests	Synthetic end-to-end run and 17 passing unit tests

The tests cover scoring, stale signals, catch-all results, ambiguous names, DNS caching, exhausted quota, denied approval, mismatched approval hashes, and approval reuse. They do not establish live provider availability or production readiness.

Emails Checker integration

I added [Emails Checker](#) as a manual free pre-check. Its public page supports up to 5,000 addresses, checks syntax and MX, exports Excel, and requires CAPTCHA. It explicitly says those checks do not prove that a specific inbox exists. ^[34]

The workflow therefore separates:

Stage	Result
Local syntax and DNS checks	Candidate is plausibly formatted and its domain has mail-routing evidence
Free Emails Checker handoff	Human submits the candidate file for syntax/MX screening
Optional mailbox verifier	Approved provider returns valid, invalid, accept-all, unknown, or another supported status

The registered Emails Checker platform advertises full verification, API access, and 100 signup credits. I did not invent an API endpoint or automate its CAPTCHA-protected public form. ^[34]

PAL and protected workflow

The package now uses the sequence documented in your project: **Parse** → **Ambiguity Scan** → **Latent Intent** → **Expand** → **Compile**. This replaces the provisional Extract → Enhance → Execute mapping from the earlier draft. ^[35]

The prospecting-specific implementation is:

1. Parse the offer, ICP, geography, roles, signals, quantity, exclusions, and constraints.
2. Scan ambiguities; ask one consequential question or record reversible defaults.
3. Record latent business intent without inferring permission, consent, or extra spending.
4. Expand the goal into signal hypotheses, queries, role searches, evidence requirements, and scoring.
5. Compile a bounded execution plan with tools, limits, outputs, and approval gates.

Then execute:

```
Search signals
→ Resolve companies
→ Resolve current contacts
→ Research email evidence
→ Generate labeled candidates
→ Check DNS
→ Optional approved verification
→ Score
→ Export locally
```

Protection mechanisms include source-content isolation, query and time limits, suppression screening, exact-payload verification approval, single-use approval checks, and separate candidate/verification states. Outreach, CRM writes, purchases, CAPTCHA bypass, and automatic bulk uploads remain blocked.

Search prefers the harness. An optional DDGS fallback uses its DuckDuckGo backend—not an official DuckDuckGo API—and remains disabled until explicitly enabled after deployment permission review. The library documents that backend. ^[36]

Structured lead scoring

The rubric ranks review priority; it is not a calibrated likelihood of purchase.

Component	Maximum
Confirmed ICP fit	25
Observed signal relevance	20
Signal freshness	15
Current contact-role fit	20
Source quality	10
Email evidence	10
Total	100

Hard qualification gates override the total. For example, a lead can score 90 with strong company, signal, and contact evidence but still remain `needs_review` because its email is guessed and unverified.

A lead becomes `ready_for_review` only when the qualification gates pass and an exact-address verifier reports valid. Accept-all remains unresolved; Hunter’s documented statuses distinguish it from valid. Even a valid result does not establish ownership or consent. ^[37]

Run and integration

From the extracted package directory, run the synthetic demo:

```
python -m runtime.prospect \
  --project demo \
  --brief "Find SaaS RevOps prospects" \
  --out demo-output

python -m unittest discover -s tests -v
```

For live operation, bind a trusted adapter to your harness's PAL/model extraction, search, permitted page retrieval, and approval gateway:

```
python -m runtime.prospect \
  --adapter your_trusted_harness_adapter \
  --project PROJECT_ID \
  --brief "Find Midwest SaaS companies hiring RevOps staff" \
  --suppression suppression.json \
  --out lead-output
```

Outputs include scored JSON/CSV leads, the compiled plan, source evidence, verifier states, a checker candidate file, handoff metadata, and a run report.

The remaining deployment decisions are the actual harness adapter, permitted search fallback, mailbox-verifier account and quota, suppression source, and retention policy. No live leads, verification submissions, uploads, or external writes were performed while building this draft.

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add this: Prospector Skill - find contact emails without a data enrichment provider subscription

User provides signal or ICP data

System uses PAL to extract intent

System uses PAL outputs to search web (duck duck go or serp/web search) - to get companies

System uses linkedin search to find potential people based on title, or google maps/places for small business owner contact etc

System uses syntax finder to find and guess syntax for key people email

system sends syntax guesses to mx record lookup/check to see if working inbox

system sends unverified contacts to verifier to see if it works,

sends back spreadsheet or adds to crm

Added your workflow as **Prospector Skill v0.3.0**: find professional contact emails from signals or ICP data without requiring a data-enrichment subscription, then return a spreadsheet or perform separately approved CRM writes.

The attached package updates the skill, manifest, scoring rules, CRM-preview template, and acceptance tests. This is a specification update; the previous Python runtime still needs new Maps/Places and CRM adapters to execute those additions.

Updated prospecting workflow

Step	System behavior
1. Intake	Accept signal data, ICP criteria, target companies, geography, titles, exclusions, and desired quantity.
2. PAL	Run Parse → Ambiguity Scan → Latent Intent → Expand → Compile to produce a bounded research plan.
3. Company discovery	Use the harness's web/SERP search or an approved DuckDuckGo adapter to find matching companies.
4. Contact discovery	Find potential people by title through permitted LinkedIn search discovery and company sources, or use the local-business route.
5. Local-business discovery	Use authorized Maps/Places or web search to find establishments, websites, and business telephone numbers; research owners or managers separately. Places supplies establishment information, not guaranteed owner emails. ^[55]
6. Syntax finder	Search published business emails and company naming patterns before generating explicitly labeled guesses.
7. Domain pre-check	Check syntax and DNS/MX routing. An MX result does not mark a specific guessed inbox as working.
8. Verification	Submit eligible candidates to an available mailbox verifier after exact-payload approval and a free-quota check. Otherwise retain an explicit unverified status.
9. Scoring	Score ICP fit, targeting relevance, freshness, role fit, source quality, and email evidence. Keep score separate from verification.
10. Delivery	Export a spreadsheet by default, or preview and execute exact-approved CRM creates/updates.

For local businesses, the skill keeps named owners, general business inboxes, and business telephone contacts separate. It never turns an `info@` address into a claimed owner email.

Key additions

- Signal-driven and ICP-only modes: ICP-only research no longer requires inventing a buying event.
- Corporate and local-business routes: title-based contact research and establishment-based discovery use different evidence requirements.
- Spreadsheet or CRM output: CRM delivery includes destination resolution, duplicate checks, field-level previews, approval, and actual write receipts.
- Subscription-free requirement: no enrichment-provider subscription is required. Optional search, Places, or verification services may still have costs or quotas; Places fields have documented pricing tiers. Default spending remains \$0. ^[56]
- Protected execution: no outreach, unauthorized scraping, CAPTCHA bypass, automatic credit purchases, or silent CRM mutations.

Package and integration

The ZIP contains:

- SKILL.md
- manifest-entry.yaml
- scoring.yaml
- crm-preview.template.json
- tests.md
- README.md

Keep the existing skill ID, signal-to-email-prospecting, and replace its instructions rather than registering a duplicate.

The remaining implementation work is explicit: bind the authorized local-business search tools, add route-aware scoring to Python, and connect the CRM read → preview → approval → write → receipt flow. No live searches, verifier submissions, or CRM writes were performed for this update.

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1. [pal-website-agent.SKILL.md](#)
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